

Crystal Reports for Visual Studio .NET

Application Deployment

Overview

This document discusses how to create an application deployment project to allow you to deploy Crystal Report for Visual Studio .NET with the necessary files to run Server File Report (SFR) Web Services on a Web server.

This document is only for use with Visual Studio 2002.

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Introduction

Crystal Reports for Visual Studio .NET extends the powerful reporting capability of Crystal Reports on the Microsoft .NET platform. You can use the Crystal Report Designer in Visual Studio .NET to create a new Crystal report or modify an existing Crystal report.

You can then keep the report on a local computer, or publish it as a Report Web Service on a Web server. Depending on whether you develop a Windows or Web application, you can host the report with either the Windows Forms Viewer or the Web Forms Viewer.

When developing an application that uses the **ServerFileReportService** Web Service, the necessary Web Service (ASMX) files and configuration file need to be manually added to the deployment project.

Deploying CR for VS .NET to a Web Server

To distribute and deploy a .NET application using Crystal Reports for Visual Studio .NET (CR for VS .NET), add the following components:

- Report Files (.RPT)
- Merge Modules
- Web Service Files (.ASMX)
- Configuration Files (.CONFIG)

Report Files (.RPT)

Report files (.RPT) used by the **ServerFileReportService** Web service can be added to the deployment project or manually copied to the Web server to a location specified as the 'root directory' in the CrystalServerfile.config configuration file.

How the report files are deployed depend on the need and architect of the application.

Merge Modules

CR for VS .NET provides merge modules in setup projects to make deployment a simple task. The merge modules are installed to the following folder, by default:

C:\Program Files\Common Files\Merge Modules

There are four merge modules that need to be included in a setup project to deploy reports.

- Managed.msm
- Database_Access.msm
- Database_Access_enu.msm
- Regwiz.msm

Managed.msm is used to install all the CR for VS .NET managed components, such as

- CrystalDecisions.CrystalReports.Engine.dll
- CrystalDecisions.Web.dll
- CrystalDecisions.Windows.Forms.dll

Database_Access.msm is used to install all database drivers the report uses to connect to various types of data sources. This merge module also installs export destination and format drivers, which are used to save report in different file formats, such as RTF and HTML.

In fact, this merge module is used to install all non-managed runtime components for CR for VS .NET, including charting components.

Database_Access_enu.msm is used to install language specific (localized) components. Some of the localized charting and exporting (like PDF) components are installed by this component. Each language the product is released in will have a specific version of this merge module.

Regwiz.msm is used to configure registration information and track licensing information on the deployed computers (server or client).

When building setup projects using this merge module, it is important to provide the license key for it.

When Regwiz.msm is added to a setup project, it exposes a **LicenseKey** property that must be set before building the setup project. The 19-digit license key is emailed when the product is registered. This key is also available from the VS .NET **Help | About** menu under the installed products list.

NOTE

If the license key is not set for the **Regwiz.msm** merge module, an error will occur when building the project. If the setup is installed on a target computer without specifying the license key for the merge module, various "keycodev2.dll" errors will occur.

For more information, go to <http://support.crystaldecisions.com/kbase> and search for knowledge base articles: c2010681 and c2011205.

Web Service Files

When deploying to a Web server hosting the Server File reports and Web Services, it is necessary to add the Web Services files to the deployment project and specify which directory to install the files.

The required Web Service files for deploying to a Web server include:

- ServerFileReportManager.asmx

- ServerFileReportService.asmx
- EnterpriseReportManager.asmx
- EnterpriseReportService.asmx

Configuration Files

There are two configuration files required on the Web server when utilizing Server File Report (SFR) Web Services. These files are not included in the Crystal Reports for VS .NET merge modules. These files include:

- Web.config
- CrystalServerFile.config

Web.config – This configuration file contains a <compilation> section that adds references to all of the necessary Crystal Decisions assemblies. The Web.config file should have a <compilation/> section similar to the following:

```
<compilation debug="true">
<assemblies>
<add assembly="CrystalDecisions.CrystalReports.Engine,
Version=9.1.3300.0, Culture=neutral,
PublicKeyToken=692fbea5521e1304"/>
<add assembly="CrystalDecisions.Shared, Version=9.1.3300.0,
Culture=neutral, PublicKeyToken=692fbea5521e1304"/>
<add assembly="CrystalDecisions.ReportSource,
Version=9.1.3300.0, Culture=neutral,
PublicKeyToken=692fbea5521e1304"/>
<add assembly="CrystalDecisions.Web, Version=9.1.3300.0,
Culture=neutral, PublicKeyToken=692fbea5521e1304"/>
</assemblies>
</compilation>
```

CrystalServerFile.config – This configuration file defines the location of the reports accessibly by the ServerFileReportManager.asmx Web Service. The physical path to the folder containing the reports must be specified in this file. For example:

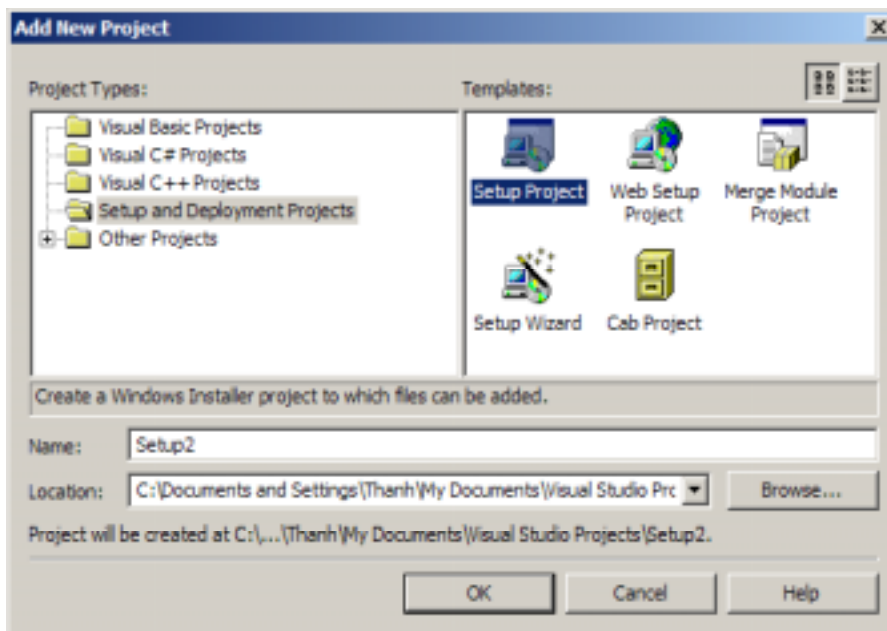
```
<RootDirectory>C:\Reports</RootDirectory>
```

Deploying a Server File Web Services Server Application

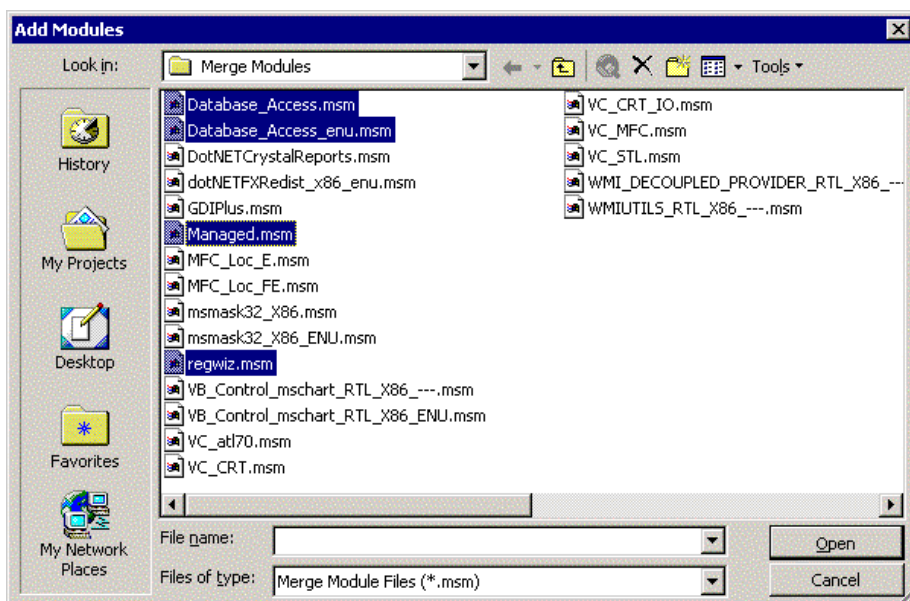
The following procedures assume that you are not deploying the Web service client with this setup package. These procedures outline how to install the CR for VS .NET components with the necessary files to a Web server such that the ServerFileReportService Web service can be consumed by a client application.

To deploy the server components for a Server File Web Service to a Web server:

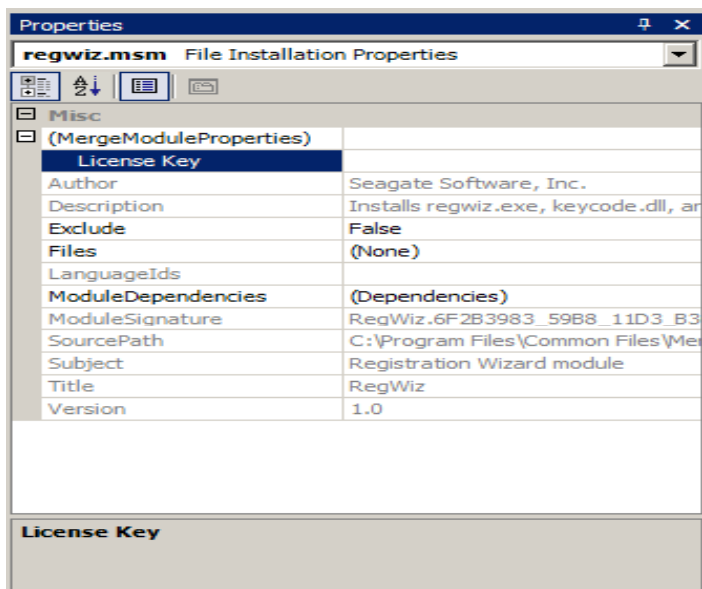
1. In VS .NET, open a new **Setup and Deployment** project and select the **Setup Project** template.



2. In the Solution Explorer, right-click the project name and select **Add** then click **Merge Module**.
3. The Add Modules dialog box appears. Select the merge modules **Database_Access.msm**, **Database_Access_enu.msm**, **regwiz.msm** and **Managed.msm**. Click OK.



4. From your Setup project, select **Regwiz.msm** to display the regwiz properties.



5. Expand the **MergeModuleProperties** and enter a valid license key the in **License Key** Properties box. The license key is the alphanumeric string you receive when registering Crystal Reports. Please note that this is **not** the 10-digit registration number.

This is **mandatory** whenever you deploy a Crystal Reports for Visual Studio .NET application.

NOTE	A license key will be given to you once you have registered Crystal Reports for Visual Studio .NET. You can also find the license key by opening VS .NET and going to the 'Help' menu and selecting 'About...'. Under the 'Installed application list' is the license key for Crystal Reports for Visual Studio .NET.
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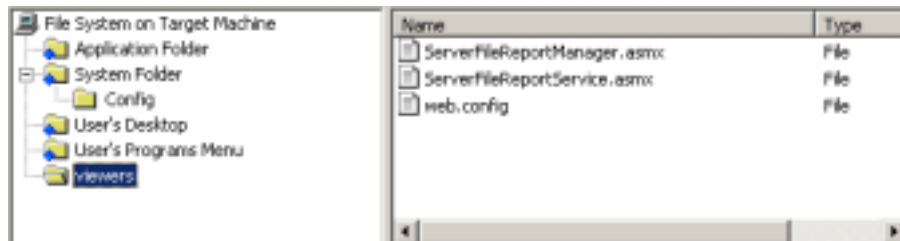
6. In the left pane of the File system tab, right-click the **File System on Target Machine** node and select **Add Special Folder...** and click **System Folder**.

Right-click the **System Folder** and select **Add** then click **Folder**. Name this folder **"Config"**.

7. Right-click the **Config** folder and select **Add** then click **File**. Add the **CrystalServerfile.config** file from the "C:\Program Files\Microsoft Visual Studio .NET\Crystal Reports\Config\" folder.

Failure to complete this step will result in "Access Denied" errors when trying to view a report through the ServerFilerRportService web service on the web server.

8. Right-click the **File System on Target Machine** node and select **Add Special Folder...** then click **Custom Folder**. Name this folder “**Viewers**”.



9. Right-click the **Viewers** folder and select **Add**. Click **File** and browse to the “C:\Program Files\Microsoft Visual Studio .NET\Crystal Reports\Viewers” folder. Add the following files:
- ServerFileReportManager.asmx
 - ServerFileReportService.asmx
 - Web.config
 - EnterpriseReportService.asmx (optional: add if using Crystal Enterprise web services)
 - EnterpriseReportManager.asmx (optional: add if using Crystal Enterprise web services)
10. With the **Viewers** folder selected, go to the Properties window and set the **DefaultLocation** property to the following path:
- [ProgramFilesFolder]\Microsoft Visual Studio .NET\Crystal Reports\Viewers**
11. To build your Setup application, right-click your deployment project in the Solution Explorer, and select **Build**.

You are now ready to deploy your Web services Server Install application.

Deploying to a Client/Server Computer

When deploying to a server computer, ensure that the computer already has the .NET Framework installed. The .NET Framework can be found on the Visual Studio 7 Windows Components Update CD or from the Microsoft’s web site at www.microsoft.com.

This document does not cover deploying the Web Service client application. If the Web Service client application is a Windows application, follow the steps outlined in the white paper, **crnet_deployment.pdf**. Download this white paper from:

<http://support.crystaldecisions.com/docs>

If the Web Service client application is a Web application and is to be installed on the same Web server as the Web Service, you can add the Setup and Deployment project described in this paper to the solution containing the Web

project and add the Web Project to the Setup Project as Project output as described in the **crnet_deployment.pdf**.

Additional Resources

For additional information on how to create an application using Server File Report Web Services download the following:

- Go to <http://support.crystaldecisions.com/docs> and search for:
 - Crnet_server_file_report.pdf
- Go to <http://support.crystaldecisions.com/downloads> and search for:
 - Csharp_win_ServerFileReports.exe
 - Csharp_win_ServerFileReports.exe
 - Vbnet_win_ServerFileReports.exe

Contacting Crystal Decisions for Technical Support

We recommend that you refer to the product documentation and that you visit our Technical Support web site for more resources.

Self-serve Support:

<http://support.crystaldecisions.com/>

Email Support:

<http://support.crystaldecisions.com/support/answers.asp>

Telephone Support:

<http://www.crystaldecisions.com/contact/support.asp>